

Week 4 Make controls reliable · paper route answers

Teacher only. Keep this sheet away from pupil copies.

GRID 1, green flag. Model answer: row 1 is go to x -180 y -120. A pupil's own start is correct instead, as long as the square is not grey and the same pair appears in the My safe start box. Rows 2 to 4 stay empty. Crossing spare rows out is correct, not a mistake.

GRID 2, right arrow. Row 1 is change x by 10. Nothing else.

GRID 3, left arrow. Row 1 is change x by -10. Nothing else.

GRID 4, up arrow. Row 1 is change y by 10. Nothing else. This is the script the fault card breaks.

GRID 5, down arrow. Row 1 is change y by -10. Nothing else.

GRIDS, non-writing pupils. Correct looks like this: the pupil points to one block in the word bank, an adult writes it into row 1 and reads it back, and the pupil says or signals yes. Correct also looks like the pupil drawing an arrow in the row and saying the axis. The choice must be the pupil's. If an adult names the block first, that row is not evidence of creation.

MAZE, safe start. Any square with x from -190 to 190 and y from -130 to 130 that is not shaded grey. Not grey means not on wall 5 (x -70, -60 or -50 with y from -130 up to 80) and not on wall 6 (x 50, 60 or 70 with y from -80 up to 130). The model start -180, -120 is always acceptable.

MAZE, single presses from the model start. Right lands on -170, -120. Left lands on -190, -120. Up lands on -180, -110. Down lands on -180, -130. All four squares are clear of every wall.

MAZE, walls. Moving the counter through a wall is expected at this stage. Do not mark it wrong. The collision consequence is the Week 5 task.

TABLE 1, predict. right arrow: moves right, new x -170, new y -120. left arrow: moves left, -190 and -120. up arrow: moves up, -180 and -110. down arrow: moves down, -180 and -130. A prediction that turns out wrong is still a completed prediction. Mark the prediction as done, then let the trace correct it.

TABLE 2, Trace 1. Step 2 is -170 and -120. Step 4 is -190 and -120. Step 6 is -180 and -110. Step 8 is -180 and -130. Steps 1, 3, 5 and 7 are printed already and must not be changed.

TABLE 3, Trace 2, the return. Step 2 is -170, -120, not the same as the start. Step 3 is -170, -110, not the same. Step 4 is -180, -110, not the same. Step 5 is -180, -120, the same as the start. This table is the practical demonstration. It comes out right only when all four step sizes are 10.

TABLE 4, Trace 3, own start. Call the pupil's own start x and y. Step 2 is x plus 10, y. Step 3 is x plus 10, y plus 10. Step 4 is x, y plus 10. Step 5 is x, y, the same as step 1. Tick the check row when step 5 matches step 1.

TABLE 5, fault card. Row 1, change x by 10, gives -170 and -120: Player moves right. Row 2, change y by 10, gives -180 and -110: Player moves up. The pupil circles row 2. The expected explanation is that the broken script changes x, which is across, and

up needs y. Changing only the sign, to change x by -10, still uses the wrong axis and moves left. Accept the pupil pointing at the letter x and shaking their head, or saying wrong letter, or drawing a sideways arrow beside row 1.

TABLE 6, which has. Line 1 is when green flag clicked. Line 2 is when right arrow key pressed. Line 3 is when left arrow key pressed. Line 4 is when up arrow key pressed. Line 5 is when down arrow key pressed. Saying the key name alone, right, is correct for a non-writing pupil. This table is the record that the pupil chose the event, not only the motion block.

TABLE 7, challenge. Row 1: -170, -120, moves right. Row 2: -170, -120, still moves right. Facing does not change what change x by 10 does. Row 3: -180, -110, moves up, because move 10 steps follows the arrow. The point to draw out is that x and y controls work the same whichever way Player faces. The final tick confirms the pupil kept change x and change y in the four scripts and did not swap in move 10 steps.

POINTING, DRAWING AND SAYING, in general. A pointed answer is correct when the pupil points unaided and an adult reads it back and gets a yes. A drawn answer is correct when the arrow direction matches the coordinates written or spoken. A said answer is correct when the words are recorded as spoken, even if they are not the pack's words: moves across, plus ten, and the sideways one are all acceptable for change x by 10. Write the exact words.

WHAT DOES NOT COUNT. A grid copied from this answer page. A block named for the pupil before they chose it. A trace filled in without the counter being moved. Any of these means the row is not observed, and it must be re-checked by having the pupil build the script again on a fresh grid.

TIMING, the same as the screen route. Making and writing the scripts 25 to 32 minutes. Individual demonstration and pupil voice 32 to 35 minutes. Exit 35 to 40 minutes.

FOLDER STEP, the paper Exit. Write the name, the date and W04 Controls at the top of the sheets. Put them in the school folder. Take them out again at 35 to 40 minutes. Check that all five grids and all the traces are there. Ask the pupil to repeat one key test from their own writing. That is the paper form of save, reopen and test, and it is observed, not assumed.

MISCONCEPTION TO WATCH. Up uses positive y in Scratch. A negative change is a direction, not an error. The control scripts belong on Player, not on Maze or Finish.

What this week evidences

AQA 71638 outcome 2: Observe pupil creation of all four arrow scripts and a practical demonstration. Check any previously inherited controls by having the pupil construct the relevant script.

Known limit of the paper route

One thing does not transfer, and it is the pack's fourth check: "I saved, reopened and tested my controls." Half of that check is a file operation - File then Save to your computer, a file named W04_Controls.sb3, a school folder on a school system. On paper no .sb3 exists, so that half cannot be evidenced and must not be ticked as though it were. The other half - that the work persists, is reopened and still runs - is evidenced and observed: the sheets go into the pupil's folder, come back out at 35 to 40 minutes, and the pupil repeats one key test from their own writing, unaided. AQA 71638 outcome 2 as published (observe pupil creation of all four arrow scripts and a

practical demonstration) is carried whole by paper, because it asks for creation and for demonstration and never asks for a file; the evidence AQA requires is a summary sheet completed by the teacher from individual observation, and every one of those observations is available here. So the named gap is the file-handling wording inside check 4, not the outcome. The teacher must record in the teacher guide, not on the AQA summary sheet, that the route was paper and that check 4 was met as folder, reopen and retest. Two smaller things, said plainly. First, the hat block is printed at the top of each grid, so the pupil chooses the motion block and the numbers but does not produce the event block from an empty page. The Which hat block starts it? table exists to close that, and the teacher should observe the pupil name the key rather than read it off the grid; if a pupil only copies the printed hat, that part is not evidenced. Second, on paper a wall cannot physically stop the counter, so a pupil can trace a move straight through a wall and still be marked correct - but that is equally true on screen in Week 4, because collision is the Week 5 task, so it is not a weakness of the paper route.